VINOD KUMAR VEMULA

AI Engineer | ML Engineer

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PERSONAL STATEMENT

AI/ML Engineer with 7 years of experience developing production-grade AI/LLM systems, scalable backend services, and real-time data platforms using Python, FastAPI, Kubernetes, Docker, and major cloud platforms including AWS, Azure, and GCP. Experienced in LLM deployment, inference optimization, cloud-native APIs, and distributed AI infrastructure.

TECHNICAL SKILLS

GenAI	LangChain, RAG Pipelines, Vector Databases, Semantic Search, Prompt Engineering, AI Observability
Programming	Python (FastAPI, asyncio, Pydantic, pytest), Core Java, SQL
Cloud Platforms	AWS (S3, Lambda, Glue, Redshift, EKS), Azure (Azure ML, Synapse, ADF, Azure DevOps), GCP (Vertex AI, BigQuery, Dataflow, Pub/Sub)
ML/AI Frameworks	TensorFlow, PyTorch, scikit-learn, XGBoost, MLflow, Model Training, MLOps
Data Engineering	Apache Kafka, Apache Spark, Snowflake, Apache Airflow, DBT
DevOps & Infrastructure	Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, CloudFormation, Logging, Monitoring & Alerting
Visualization	Power BI, Tableau, Grafana

PROFESSIONAL EXPERIENCE

Gen AI Engineer |CGI **|** Oct 2025 – Jun 2026 **|** Dallas, TX

Building and deploying enterprise AI and LLM applications using Python, FastAPI, Vertex AI, and Gemini to support scalable conversational AI and intelligent automation workflows.

- Built scalable AI/ML deployment pipelines with automated CI/CD workflows, enabling controlled releases, version management, monitoring, and rollback capabilities for production AI systems supporting insurance operations.
- Designed and optimized asynchronous LLM inference APIs using Python asyncio, caching, and request orchestration, reducing average response latency by nearly 40% under concurrent workloads and improving responsiveness for business-critical insurance workflows.
- Implemented AI observability frameworks to monitor latency, token consumption, inference failures, response quality, and service reliability, helping maintain consistent AI performance for claims processing and operational support applications.
- Containerized AI services using Docker and supported deployments in Kubernetes/EKS environments, implementing health checks, autoscaling, and resource optimization to improve service availability and handle fluctuating enterprise workloads.
- Deployed and managed production-grade LLM services using Python, FastAPI, Vertex AI, Gemini, Docker, and Kubernetes, enabling reliable AI-powered automation for insurance business processes.
- Developed RAG-based enterprise AI pipelines integrating embeddings, semantic search, vector retrieval, and internal knowledge systems, helping Claims and Operations teams retrieve relevant policy information, procedural guidelines, and business documentation more efficiently.
- Optimized AI inference environments by resolving deployment bottlenecks, improving resource utilization, and stabilizing distributed LLM workloads to support dependable, low-latency responses for enterprise users.
- Developed secure backend services using FastAPI, Pydantic, and REST APIs to integrate enterprise applications with LLM and ML inference workflows, enabling AI-assisted information retrieval, workflow automation, and operational decision support.

AI/ ML Data Engineer |BMO Bank **|** Jun 2023 – Sep 2025 **|** Chicago, IL

Built ML-driven fraud detection, risk-scoring, and real-time analytics platforms within a regulated banking environment.

- Developed and deployed ML-based fraud detection and risk-scoring services using Python, FastAPI, Azure ML, Kafka, and Spark Streaming for real-time financial transaction analysis.
- Built and maintained real-time streaming pipelines processing high-volume financial events with focus on low latency, reliability, and production stability.
- Exposed ML inference services through REST APIs enabling downstream applications to consume fraud risk predictions in real time.

- Implemented CI/CD deployment pipelines for ML workloads using Azure DevOps, automating testing, deployment validation, and release management.
- Monitored model-serving performance, streaming job health, API latency, and infrastructure reliability across production financial systems.

Cloud & ML Engineer – Internship |John Deere **|** Aug 2022 – Jan 2023 **|** Moline, IL

Built ML-ready cloud data pipelines for IoT sensor analytics supporting predictive maintenance and anomaly detection across 10,000+ connected agricultural machines.

- Engineered AWS-based data pipelines using S3, Lambda, and Redshift to ingest and process IoT telemetry streams, generating structured feature datasets for predictive maintenance and anomaly detection models.
- Automated infrastructure provisioning with Terraform and CloudFormation, cutting setup time by 25% and reducing infrastructure costs by 20%.
- Implemented sensor data validation and quality checks across IoT ingestion pipelines, ensuring reliable ML training datasets and maintaining 99.8% pipeline availability.

Data Platform Engineer | Infosys **|** Mar 2019 – Dec 2021 **|** India

Built real-time data processing pipelines and cloud-based analytics workflows for consumer product analytics using Google Cloud Platform, Kafka, and distributed data engineering technologies.

- Automated CI/CD workflows with Cloud Composer, reducing deployment errors by 15%.
- Implemented Cloud Pub/Sub and Kafka for real-time analytics processing 500K+ daily transactions.
- Developed Python and SQL-based transformation pipelines to process product sales and customer behavior datasets supporting analytics across 50+ consumer products.

Data Engineer | Aviva Life Insurance **|** Jan 2018 – Feb 2019 **|** Bangalore, India

Built cloud-based ETL and analytics pipelines supporting insurance policy processing, claims analytics, and regulatory reporting across enterprise-scale datasets.

- Developed and optimized BigQuery ETL pipelines processing 500K+ insurance policy records, ensuring 99.9% pipeline uptime and reliable regulatory data delivery across reporting systems.
- Integrated AWS S3 and Redshift with GCP-based data processing workflows, enabling efficient multi-cloud data movement and reducing enterprise data transfer costs by 15%.
- Utilized Sqoop and Impala for high-speed ingestion and analytics of large-scale claims datasets, supporting regulatory reporting, policy auditing, and enterprise compliance operations.

PROJECTS

DeFi Protocol Monitor **Apr 2026- Present**

- Built production-grade data engineering pipeline using FastAPI, Python, and Docker to monitor DeFi protocols (HyperEVM). Implemented real-time anomaly detection for TVL drops (20% threshold), APY changes (2% minimum), and utilization spikes (95% threshold).
- Features REST API endpoints for health metrics and alert history with SQLite database backend.

Multimodal AI Copilot System **Mar 2026 – Apr 2026**

- Developed a multimodal AI platform capable of processing audio and visual streams to generate real-time insights and automated incident reports using LLM-based reasoning workflows.
- Designed event-driven backend pipelines and low-latency inference orchestration to support scalable AI processing in high-throughput environments.

University Hybrid Chatbot (RAG System) **Mar 2025 – May 2025**

- Built a Retrieval-Augmented Generation (RAG) chatbot using Azure OpenAI, FastAPI, Docker, and vector-based retrieval workflows to provide contextual responses for university admissions and course-related queries.
- Developed backend APIs, semantic search integration, and cloud deployment workflows using Azure App Services.

CERTIFICATIONS

Certified GenAI Expert| AI Capstone Project with Deep Learning | Introduction to Neural Networks and PyTorch | AWS Data Engineer |GCP Professional Data Engineer

EDUCATION

Master of Science in Statistics and Decision Analytics Western Illinois University, USA **|** 2022 - 2023 | 3.60

Bachelor of Technology in Mechanical Engineering Andhra University, India **|** 2014 - 2018 | 3.00